Loan Approval Prediction Using Machine Learning

Introduction

Loan approval is an important process for financial institutions, as it involves evaluating an applicant's ability to repay a loan. Traditional loan approval methods rely on manual evaluation, which can be time-consuming and influenced by human bias. This project aims to build a machine learning model to predict whether a loan application will be approved or rejected using applicant financial and demographic data.

Dataset Description

The dataset consists of information related to loan applicants, including income, loan amount, loan term, credit score (CIBIL score), number of dependents, employment status, education level, and asset values. The target variable is **loan_status**, which indicates whether a loan was approved or rejected. The **loan_id** column was removed as it does not contribute to prediction.

Data Preprocessing

The data was cleaned by standardizing column names and handling missing values. Numerical features were filled using median values, while categorical features were filled with the most frequent values. Categorical variables were encoded, and numerical features were scaled to ensure consistent model performance. The dataset was then split into training and testing sets.

Methodology

This project was treated as a binary classification problem. Two models were used: Logistic Regression as a baseline model and Random Forest as an advanced model to capture complex patterns. Both models were trained on the same dataset and evaluated using standard classification metrics.

Method-1 Logistic Regression:

[11]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

[12]:
```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

[13]:
```
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

print("Logistic Regression Performance")
print(classification_report(y_test, y_pred_lr))
```

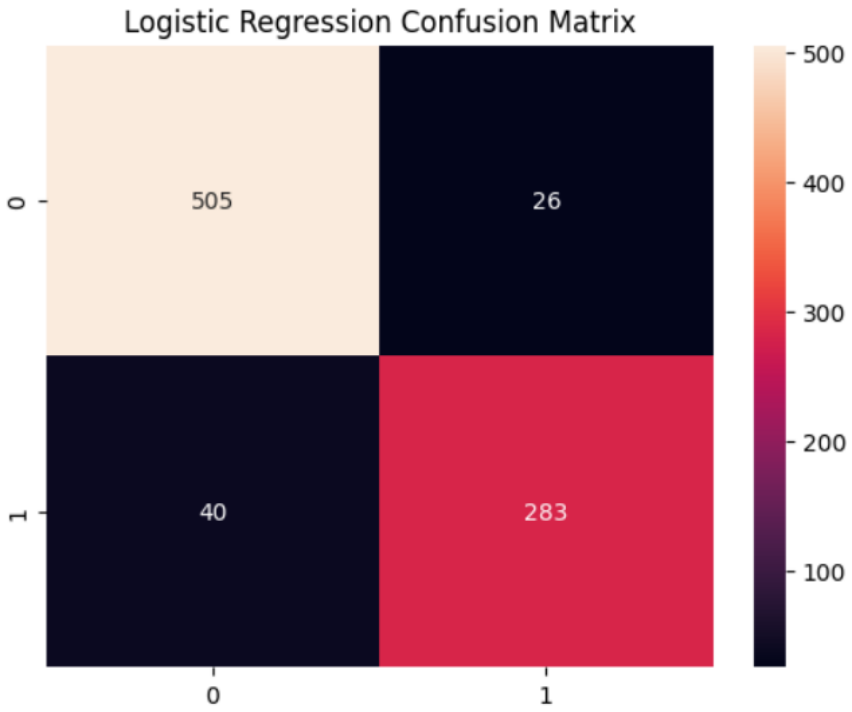
```
Logistic Regression Performance
              precision    recall  f1-score   support

    Approved       0.93      0.95      0.94       531
    Rejected       0.92      0.88      0.90       323

    accuracy                           0.92       854
   macro avg       0.92      0.91      0.92       854
weighted avg       0.92      0.92      0.92       854
```

Confusion Matrix:

```
[14]: sns.heatmap(confusion_matrix(y_test, y_pred_lr), annot=True, fmt='d')
plt.title("Logistic Regression Confusion Matrix")
plt.show()
```



Method-2 Random Forest:

```
[16]: rf = RandomForestClassifier(n_estimators=100, random_state=42)
      rf.fit(X_train, y_train)

      y_pred_rf = rf.predict(X_test)

      print("Random Forest Performance")
      print(classification_report(y_test, y_pred_rf))
```

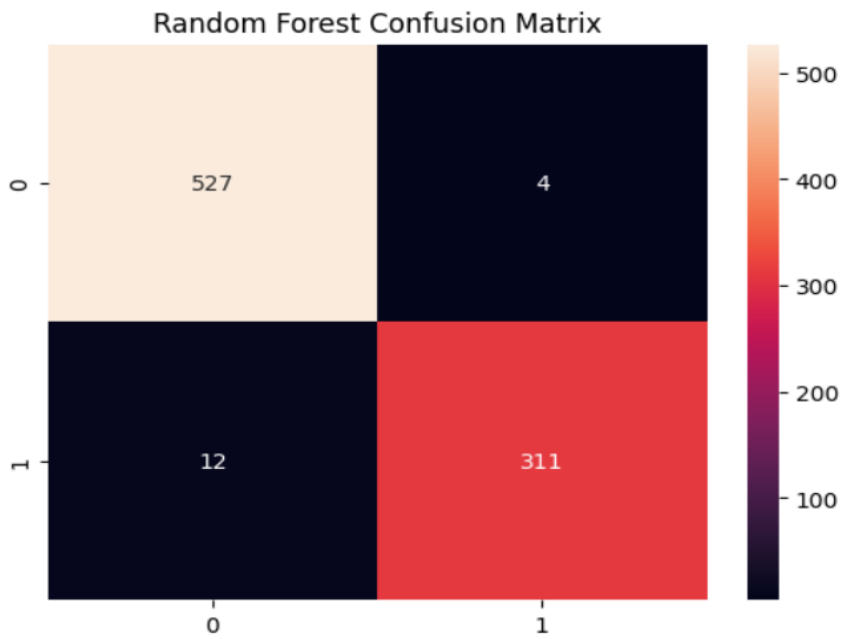
```
Random Forest Performance
              precision    recall  f1-score   support

    Approved       0.98      0.99      0.99       531
    Rejected       0.99      0.96      0.97       323

    accuracy                           0.98       854
   macro avg       0.98      0.98      0.98       854
weighted avg       0.98      0.98      0.98       854
```

Confusion Matrix:

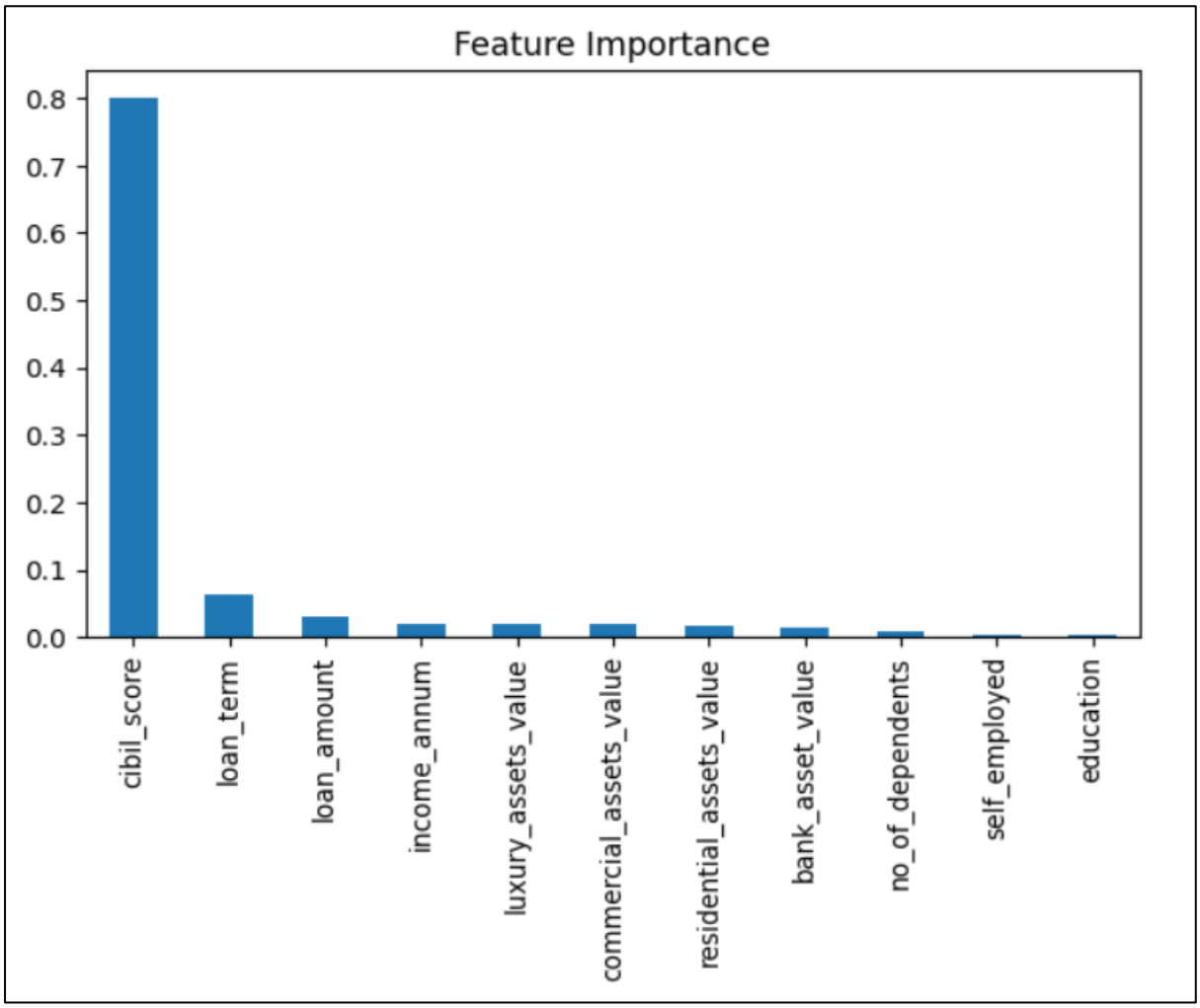
```
[17]: sns.heatmap(confusion_matrix(y_test, y_pred_rf), annot=True, fmt='d')
      plt.title("Random Forest Confusion Matrix")
      plt.show()
```



Feature Importance:

```
[20]: feature_importance = pd.Series(
          rf.feature_importances_,
          index=X.columns
      ).sort_values(ascending=False)

      plt.figure(figsize=(7,4))
      feature_importance.plot(kind='bar')
      plt.title("Feature Importance")
      plt.show()
```



Model Comparison:

- Logistic Regression provides an interpretable baseline.
- Random Forest captures non-linear relationships and performs better.
- Credit score and income-related features are the most influential.

Results and Discussion:

The Random Forest model performed better than Logistic Regression, showing improved accuracy and overall prediction performance. Feature importance analysis indicated that credit score, annual income, and asset values were the most influential factors in loan approval decisions. The results show that machine learning can effectively support automated and consistent loan approval decisions.

Conclusion:

This project demonstrates how machine learning can be used to automate loan approval decisions. The Random Forest model outperformed Logistic Regression and successfully identified key financial factors influencing loan approval decisions.